



STQD6014 Data Science – Project Report 2

Integrated Analysis of State Real GDP and Labour Force (Malaysia, 2023)

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1. Introduction

This report integrates two datasets from the Department of Statistics Malaysia (DOSM) to examine how state-level economic output and structure relate to labour market outcomes in 2023. The first dataset, Annual Real GDP by State and Economic Sector (Supply Approach), provides annual state gross domestic product (GDP) at constant 2015 prices (million RM) and decomposes GDP into sector components. The second dataset, Annual Principal Labour Force Statistics by State and Sex, provides labour force counts and key rates including labour force participation rate and unemployment rate disaggregated by sex (male, female, and total).

The primary objective is to merge the two datasets at the state level for the year 2023 and to investigate relationships between GDP structure and labour market outcomes. The work also demonstrates an applied data integration workflow: cleaning and reshaping the data, harmonising join keys, validating the merge, and communicating insights through descriptive statistics and visualisations.

1.1 Dataset Descriptions

GDP dataset. The GDP data are in long (tidy) format with columns *series*, *date*, *state*, *sector*, and *value*. The *series* column distinguishes absolute values (*abs*) from year-on-year growth rates (*growth_yoy*). The *date* column stores the year as a YYYY-01-01 timestamp. Sector codes are mapped as follows: p1 = Agriculture; p2 = Mining and Quarrying; p3 = Manufacturing; p4 = Construction; p5 = Services; p6 = Taxes less subsidies on products. GDP values are recorded in million RM at constant 2015 prices.

Labour force dataset. This dataset contains state, sex, date, labour force counts (*lf*, *lf_employed*, *lf_unemployed*, *lf_outside*) and rates (*p_rate*, *u_rate*, *ep_ratio*). Counts are

recorded in thousands of persons and rates are percentages. In 2023, the dataset includes totals (sex = both) for all 16 states and territories, as well as male and female breakdowns for most — but not all — states.

2. Data Preprocessing and Integration

Both datasets were filtered to the year 2023. A check confirmed that each dataset contains records for 16 states and territories. A key name mismatch was identified: the labour dataset refers to the federal capital as "W.P. Putrajaya", while the GDP dataset uses the label "Supra" for a national aggregate category. To enable a valid join, the GDP value "Supra" was replaced with "W.P. Putrajaya". This record is retained in the merged dataset but flagged as anomalous and excluded from sector-share and productivity scatter plots to prevent one questionable record from distorting conclusions.

The 2023 GDP data were reshaped from long to wide format so that each state occupies one row, with columns for total GDP (p0) and each sector value (p1–p6). The proportion of each sector component relative to total GDP (p0) was calculated to enable meaningful comparisons of economic structure across states that differ greatly in absolute GDP size. The labour dataset was similarly reshaped to produce separate columns for total, male, and female values and rates. For Negeri Sembilan, male and female rows are absent from the 2023 labour data; the state is retained for overall statistics but excluded from sex-specific visualisations.

After cleaning, both datasets were merged on state using an inner join, producing an integrated 2023 table with 16 rows and columns from both sources. GDP per employed person was then computed by dividing total GDP (million RM) by employed persons (thousands) and converting the result to RM per person.

3. Analysis and Findings

3.1 State GDP Rankings

In 2023, Selangor recorded the largest real GDP (approximately RM 406,071 million), followed by W.P. Kuala Lumpur (approximately RM 249,302 million) and Johor (approximately RM 148,205 million). Sarawak (approximately RM 142,352 million) and Pulau Pinang (approximately RM 115,957 million) complete the top five. GDP is therefore heavily concentrated in the Klang Valley corridor and the most industrialised states.

3.2 Sector Composition by State

Figure 1 shows the proportion of total real GDP accounted for by each sector across all 16 states, ranked by total GDP. Services dominate in most states, particularly in W.P. Kuala Lumpur (over 90% of GDP), reflecting its role as a centre for finance, business services, trade, and public administration. Manufacturing is most prominent in Pulau Pinang (approximately 46% of GDP), with high shares also in Negeri Sembilan (approximately 38%) and Terengganu (approximately 37%). Mining and Quarrying accounts for around 23% of GDP in Sabah and 21% in Sarawak, reflecting these states' natural resource endowments. Agriculture has the

highest shares in Pahang (approximately 22%) and Kelantan (approximately 20%), with implications for the types of employment available in those states.

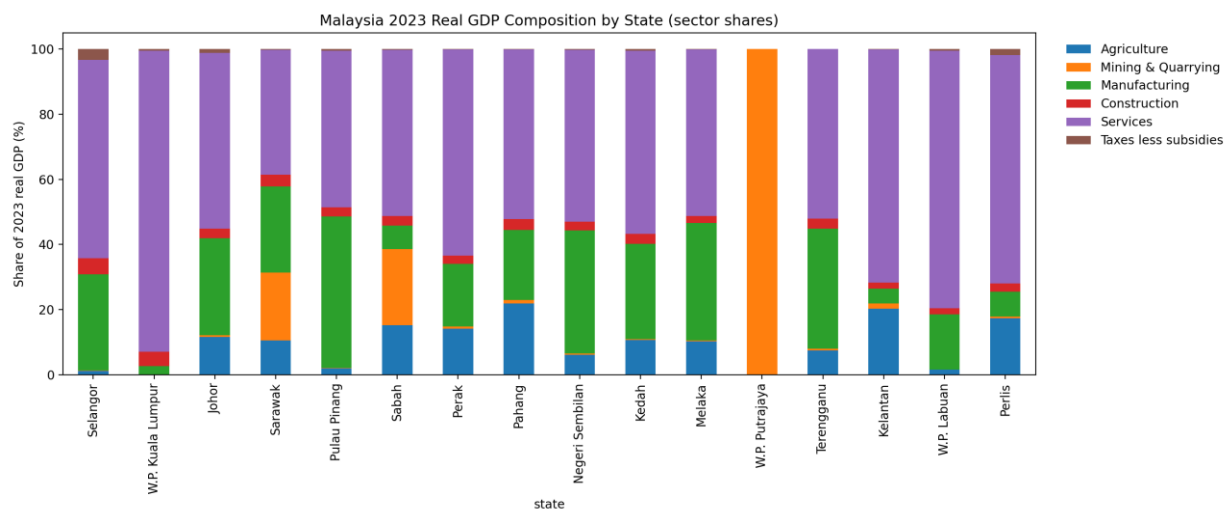


Figure 1. Malaysia 2023 Real GDP Composition by State (sector shares). States are ordered by total real GDP from largest (Selangor) to smallest (Perlis).

3.3 Labour Outcomes, Unemployment Hotspots, and Gender Gaps

Unemployment in 2023 was highest in Sabah (7.5%) and W.P. Labuan (6.8%), while Melaka (1.6%) and Pulau Pinang (2.2%) recorded much lower rates. Labour force participation was strongest in urban regions Selangor (77.4%) and W.P. Kuala Lumpur (74.2%) and fell to just 57.3% in Kelantan.

As shown in Figure 2, men participate in the labour force at substantially higher rates than women in nearly every state. The gap is especially wide in Johor (88.4% vs. 50.8%), W.P. Labuan (85.4% vs. 50.0%), and Terengganu (76.6% vs. 42.0%). Urban states show narrower gender gaps: female participation reaches 66.5% in W.P. Kuala Lumpur and 69.7% in Selangor, suggesting that service-oriented urban economies create more accessible employment opportunities for women, though cultural and structural factors also play a role.

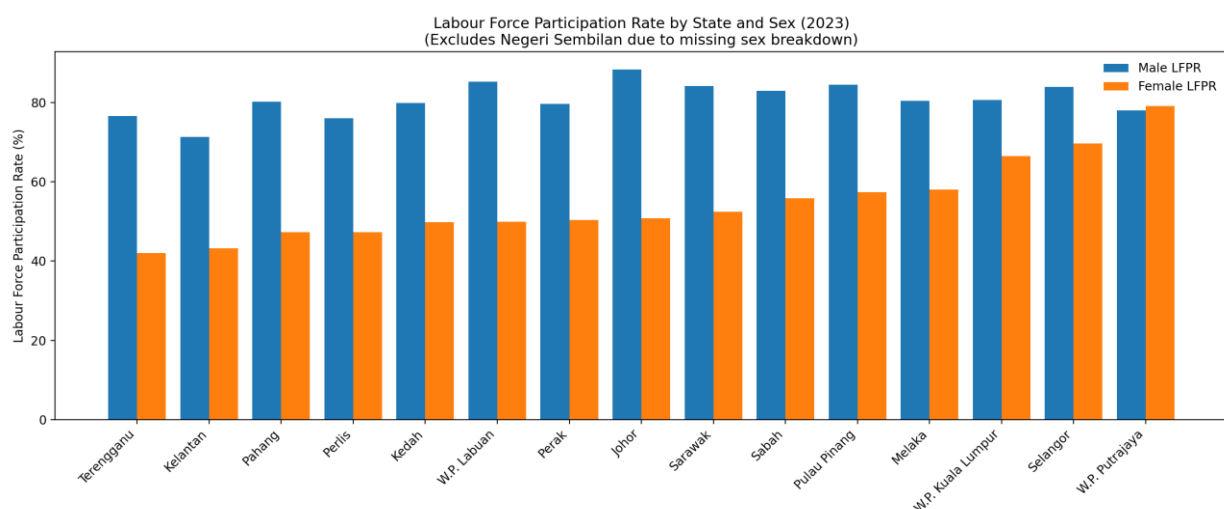


Figure 2. Labour Force Participation Rate by State and Sex (2023). Negeri Sembilan is excluded due to missing male and female breakdowns in the 2023 data.

3.4 Sector Mix and Labour Outcomes

Figure 3 plots the services sector share of GDP against female labour force participation rate across states (excluding W.P. Putrajaya, whose sector split is anomalous due to the Supra mapping). The Pearson correlation is positive but weak ($r = 0.18$), indicating that a larger services sector is associated with modestly higher female participation, but the relationship is not strong enough to be conclusive at this scale.

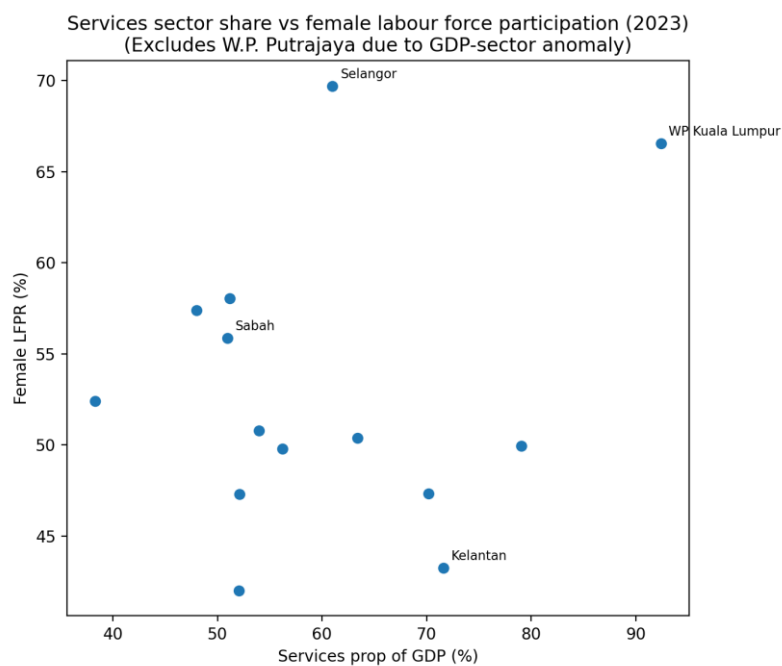


Figure 3. Services sector share of GDP vs. female labour force participation rate (2023). W.P. Putrajaya is excluded due to the GDP-sector anomaly.

Figure 4 shows GDP per employed person against the unemployment rate. W.P. Labuan stands out as a high-productivity state with an elevated unemployment rate, demonstrating that high output per worker does not necessarily translate into broad employment opportunities a pattern often seen in capital-intensive sectors such as offshore finance. Sabah combines a low GDP per employed person with the highest unemployment rate, reflecting the structural challenges of a natural-resource-dependent economy.

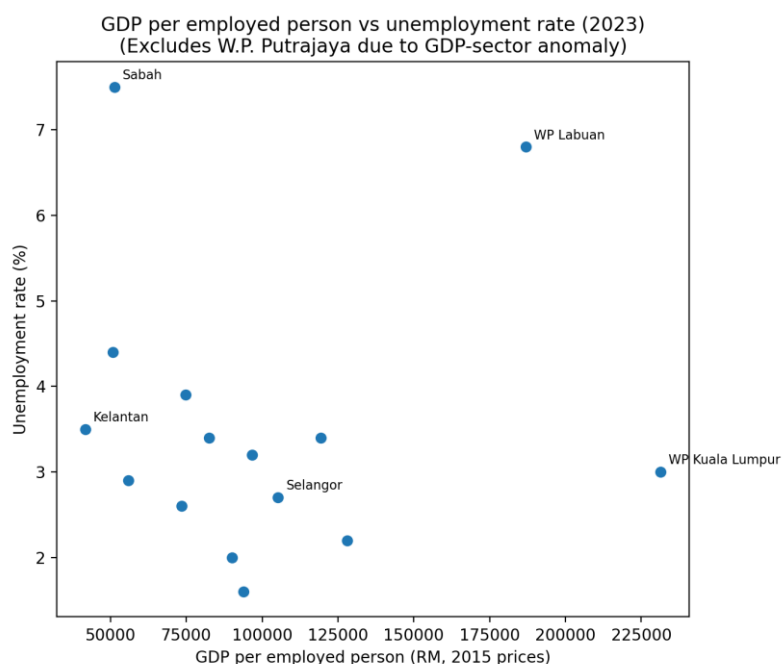


Figure 4. GDP per employed person (RM, 2015 prices) vs. unemployment rate (2023). W.P. Putrajaya is excluded.

Table 1 summarises the Pearson correlations between economic structure variables and labour market outcomes.

Variable A	Variable B	r	Strength
Services share (p5)	Female LFPR	+0.18	Weak positive
Mining share (p2)	Unemployment rate	+0.50	Moderate positive
Manufacturing share (p3)	Unemployment rate	-0.52	Moderate negative
Overall LFPR	Unemployment rate	-0.31	Weak negative

Table 1. Pearson correlations between sector composition and labour market outcomes (2023, excluding W.P. Putrajaya).

Mining's share of GDP shows a moderate positive correlation with unemployment ($r = +0.50$), consistent with the observation that Sabah a high-mining state has the highest unemployment rate. Manufacturing's share shows a moderate negative correlation ($r = -0.52$), suggesting that states with stronger manufacturing bases tend to have lower unemployment. This is consistent with manufacturing's role as a labour-intensive sector that generates broad formal employment. The weak negative correlation between overall labour force participation and unemployment ($r = -0.31$) indicates a modest tendency for higher-participation states to also have lower unemployment, though exceptions exist.

4. Challenges and How They Were Addressed

Key name mismatch (Supra / W.P. Putrajaya). The GDP dataset labels a national-level aggregate category as "Supra", while the labour dataset uses "W.P. Putrajaya" for the federal administrative capital. These two entries do not refer to the same geographic unit, but

mapping Supra to W.P. Putrajaya was necessary to achieve a 16-row join. The resulting record was flagged and excluded from all sector-share and productivity relationship plots so that this anomalous mapping does not distort cross-state comparisons.

Missing sex breakdown for Negeri Sembilan. The 2023 labour dataset does not include separate male and female rows for Negeri Sembilan. The state is retained for overall (sex = both) statistics and is present in the merged dataset, but it is excluded from Figure 2 and the services-vs-female-LFPR scatter plot (Figure 3) to avoid incorrect imputation.

5. Conclusion

By cleaning, harmonising, and merging two public DOSM datasets, this report uncovers clear regional patterns in Malaysia's economy for 2023. GDP is heavily concentrated in Selangor and W.P. Kuala Lumpur, while the mix of economic sectors varies considerably across states — from services-dominated urban centres to agriculture- and mining-dependent peripheries. Labour market outcomes differ correspondingly: participation rates and unemployment rates both show significant state-level variation, and gender gaps in participation remain large in most states, though they narrow in more urbanised, service-oriented regions.

The correlation analysis reveals that states with stronger manufacturing bases tend to have lower unemployment, while mining-dependent states face higher unemployment, highlighting that economic growth particularly in capital-intensive sectors does not automatically translate into broad employment gains. The GDP per employed person metric further illustrates this point: W.P. Labuan achieves high productivity per worker yet records elevated unemployment, while agricultural states have low productivity but comparable or lower unemployment rates.

Taken together, this integrated analysis provides a richer picture of Malaysia's regional economic landscape than either dataset alone. It demonstrates the practical value of combining publicly available data sources and applying systematic data integration workflows — including key harmonisation, long-to-wide reshaping, flagging of anomalous records, and correlation analysis — to generate actionable socio-economic insights.

References

Department of Statistics Malaysia (DOSM). (2024). *Annual Real GDP by State and Economic Sector (Supply Approach), 2015–2023*. Retrieved from <https://open.dosm.gov.my>

Department of Statistics Malaysia (DOSM). (2024). *Annual Principal Labour Force Statistics by State and Sex, 1982–2023*. Retrieved from <https://open.dosm.gov.my>